

Assignment_Terraform_2

Terraform: Dependency Management

1. How does Terraform handle dependency management?

Terraform handles dependency management by building a dependency graph from the configuration. It studies how resources refer to each other and then decides the correct order for creating, updating, or destroying them. This means Terraform usually does not need the user to manually arrange every step.

Most dependencies are detected automatically. For example, if one resource uses the output or attribute of another resource, Terraform understands that the second resource depends on the first one. During apply, Terraform creates the required resource first and then creates the resource that depends on it. During destroy, it follows the reverse order so dependent resources are removed safely.

Terraform can also handle explicit dependencies when the relationship is not obvious from the configuration. In those cases, a user can define a dependency directly so Terraform knows that one part must wait for another part. This is useful when two resources are related logically, but Terraform cannot detect the connection just from attribute references.

This dependency system helps avoid common infrastructure problems. It prevents Terraform from trying to create resources too early, deleting resources in the wrong order, or applying changes in a way that breaks connected services. In simple words, Terraform uses the dependency graph to make infrastructure changes in a safe and organized sequence.